SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID -** SIH25072
- **Problem Statement Title -**Design and Implementation of Solar-Powered dewatering in Mining Operations.
- **Theme -** Renewable / Sustainable Energy
- **PS Category-** Hardware
- **Team ID -** 61770
- **Team Name -** MiningMotors_T72

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IDEA TITLE

PRESENTING SURYADUT- THE ULTIMATE DEWATERING SOLUTION

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HIGHLIGHTED PROBLEMS

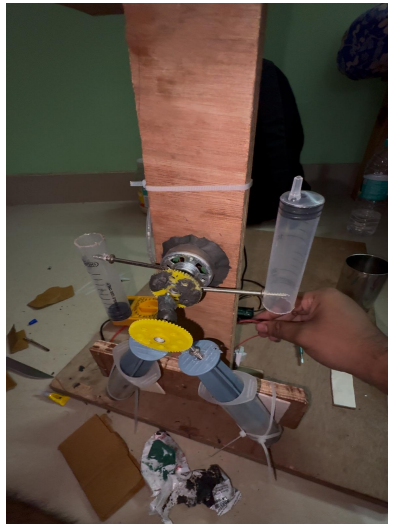
High operational costs from electricity and diesel fuel consumption
Increased carbon emissions from diesel generator dependency
Unreliable power supply affecting consistent dewatering operations
Rising energy costs impacting operational budgets
Maintenance complexity of traditional pump and generator systems
Need for sustainable mining practices to meet environmental goals
Capital expenditure constraints requiring alternative financing models

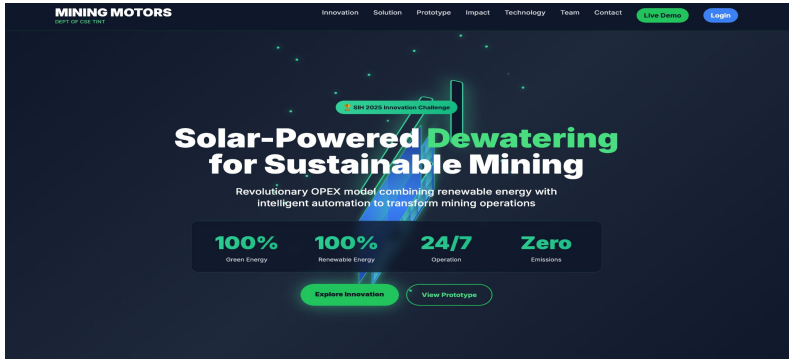
PRESENT SOLUTION

- **Escalating Energy Costs** from Grid and Diesel Usage
- Significant **Carbon Footprint** from **Diesel Backup Operations**
- **High Maintenance & Operational** Cost
- **Financial Strain** from CAPEX Model

OUR UNIQUE VALUE PROPOSITION

- **Solar-Powered Dewatering**: Low Operational Cost, Charging when working
- **Gravity + Magnetic Piston Generator** – Minimal loss of energy
- **OPEX-First Model** – When only used then only pay
- **Battery Management System** – Safety measures
- **Floating & Mobile Design** – Buoyant system with wheels allows easy relocation and stability on-site.
- **Integrated IoT Monitoring** – Real-time performance tracking and predictive maintenance via platform.
- **Minimum Maintenance Burden** – Mainly contactless mechanism & handled entirely by certified Energy-as-a-Service vendors.
- **High Energy Efficiency** – Dual magnetic pistons deliver near 360° rotation with minimal input energy.
- **Sustainability First** – Decarbonizes mine dewatering, aligning with ESG and net-zero goals.
- **Scalable & Collaborative** – Easily adaptable for mines of all sizes; open to partnerships and industry collaborations.





First Prototype demo of Suryadut & its rental portal

TECHNICAL APPROACH

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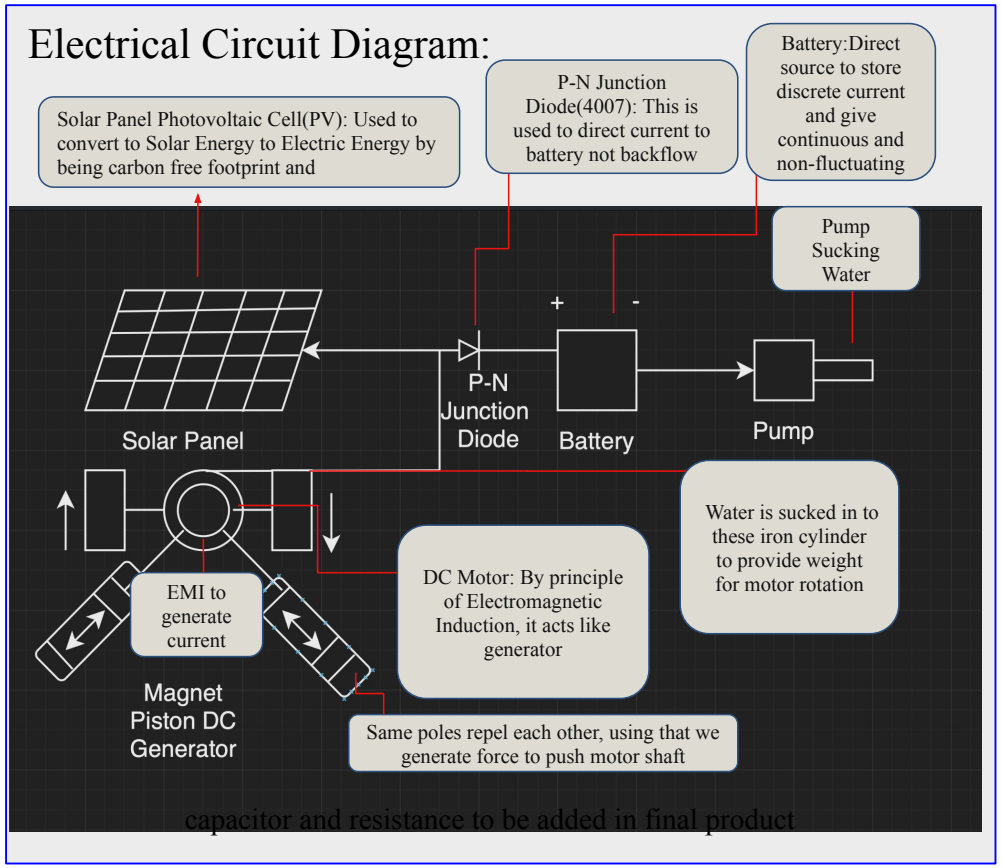
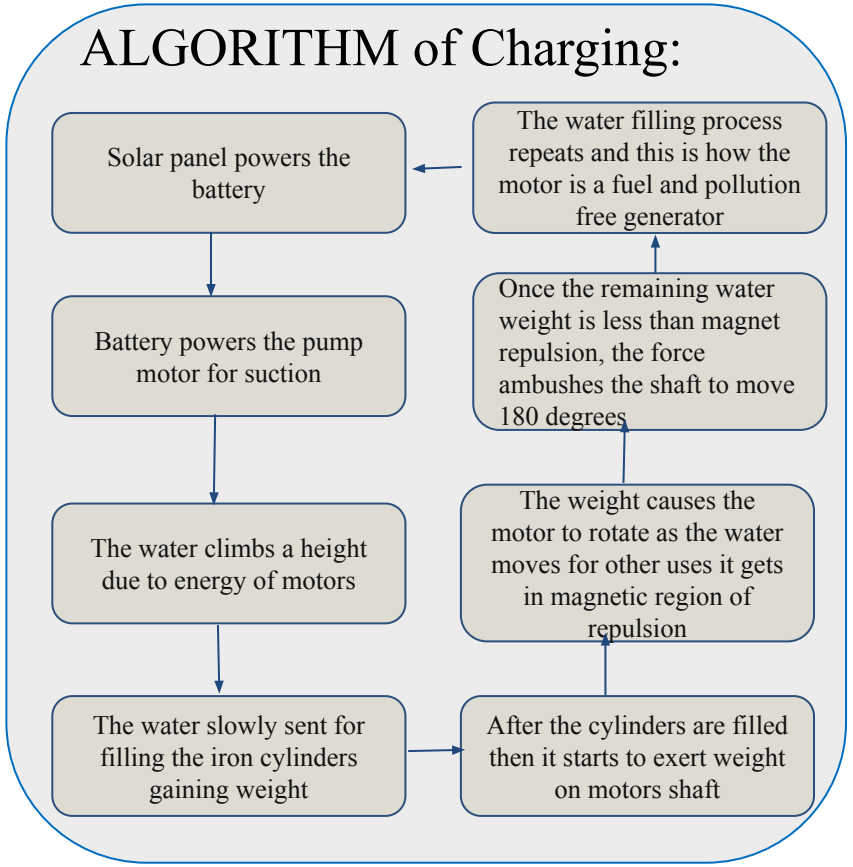
Electrical Circuit Diagram:



ALGORITHM of Charging:



Portal Tech Stack:

HTML, CSS, JS — for prototyping purpose

when upgraded to final:

- NEXT.JS — Web Development
- CSS — Advanced Designing
- Advanced Animations
- Backend logic Deployment & Server with Express
- MySQL — Database Management Systems
- REST APIs, etc.

Electronic Parts: **ESP32 controller board** as the core IoT unit with a **LoRa module** for remote connectivity. Integrate an **ultrasonic level sensor** to measure sump water levels, a **flow meter** to track discharge, and **current and voltage sensors** to monitor power consumption. Add a **temperature and vibration sensor** to assess motor health, and a **relay or contactor module** for remotely switching the pump on or off. Power the system with a **battery backup or solar panel** for reliability, and house everything in an **IP65-rated enclosure** for dust and water protection, with a **simple cloud dashboard (MQTT + Grafana)** to display data and send alerts.

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FEASIBILITY AND VIABILITY

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Technical Implementation(~80%)

- Solar panels + submersible pumps are mature, proven for remote dewatering.
- Floating barge/pontoon design widely used in open-pit mines, easy relocation, no civil works.
- IoT monitoring feasible with existing rugged controllers; improves reliability and predictive maintenance.
- Magnetic repulsion generator concept is unproven — needs testing and validation.
- Must ensure rugged, corrosion-resistant materials and sufficient solar array sizing.

Operational Feasibility(~90%)

- Modular floating systems easy to deploy and move on site; no major infrastructure needed.
- Maintenance simpler than diesel systems — mostly panel cleaning and electronics checks.
- Requires vendor support and some retraining for electrical/electronic maintenance.
- IoT telemetry reduces manual inspections and downtime.

Economic Feasibility(~75%)

- Eliminates diesel fuel cost and reduces maintenance → 25–50% operating cost savings over time.
- Higher upfront CAPEX offset by lower OPEX; ROI achievable in 3–5 years with proper sizing.
- OPEX-first or leasing model attractive for miners avoiding large capital outlay.
- Possible government/renewable subsidies in India could improve economics further.

Market Response(~85%)

- Global and Indian mining industries under pressure to cut emissions and costs.
- ESG and investor requirements drive adoption of renewable-powered systems.
- Few mining-specific solar dewatering solutions exist — early mover advantage possible.
- Demand for sustainable dewatering solutions increasing, especially in water-scarce regions.

Regulatory Compliance (India)(~90%)

- Aligns with Environment (Protection) Act, Water Act, and Mineral Conservation & Development Rules.
- Improves ESG compliance and may raise a mine's government sustainability rating.
- Benefits from India's strong policy push for renewable energy (e.g., PM-KUSUM solar pump subsidies).

Business Model

Mining Motors aims to revolutionize mine dewatering with a **solar-powered floating pump system** that cuts diesel use, lowers costs, and meets sustainability targets. Using an **OPEX-first model** (pay-as-you-go), the system combines solar panels, high-efficiency pumps, and IoT monitoring to deliver reliable, low-maintenance water removal in even the harshest mining conditions. Targeted at medium and large mines in India, the solution aligns with **environmental regulations and ESG goals**, offers rapid deployment without major infrastructure, and leverages government renewable-energy incentives to improve affordability. With growing demand for cost-efficient and eco-friendly mining solutions, Mining Motors positions itself as an early mover delivering measurable **operational savings, compliance benefits, and competitive advantage**.

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IMPACT AND BENEFITS

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Short-Term Benefits (0-2 years)

Immediate cost reduction - Lower energy costs from day one without capital investment

Risk transfer - Vendor assumes equipment performance and maintenance responsibilities

Zero upfront investment - No CAPEX burden, preserving cash flow for core mining operations

Rapid deployment - Quick system installation and commissioning within 3-6 months

Reduced diesel dependency - Immediate cut in fuel costs and supply chain risks

Operational reliability - Consistent power supply reducing downtime and maintenance issues

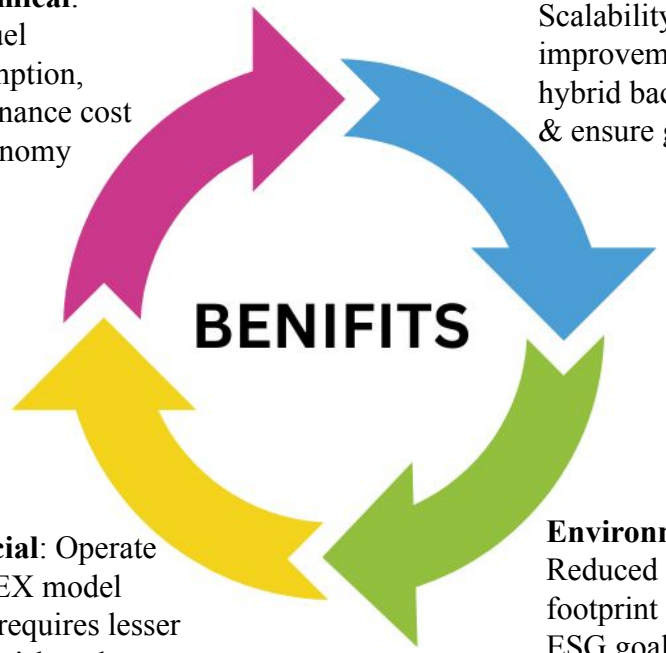
Predictable expenses - Fixed monthly payments improving budget planning and forecasting

Long-Term Benefits (3-15 years)

- **Carbon footprint reduction** -
- **Energy price stability**
- **Technology upgrades** -
- **Enhanced asset value**
- **Regulatory compliance**
- **Scalability advantages**
- **Knowledge transfer**
- **Competitive advantage**

Economical: Low fuel consumption, maintenance cost & autonomy

Sustainability: Ensures Scalability & scope for improvement ensures hybrid backup features & ensure good practices



Financial: Operate on OPEX model which requires lesser capital risk and ensuring pay as you go plans

Environmental: Reduced Carbon footprint and supports ESG goals to reach greener initiative

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RESEARCH AND REFERENCES

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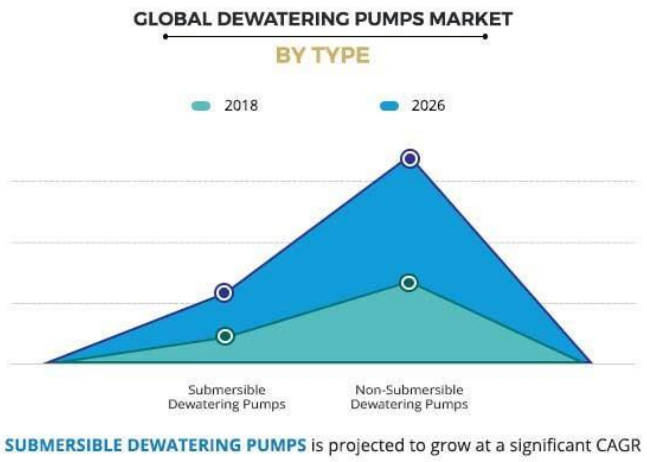
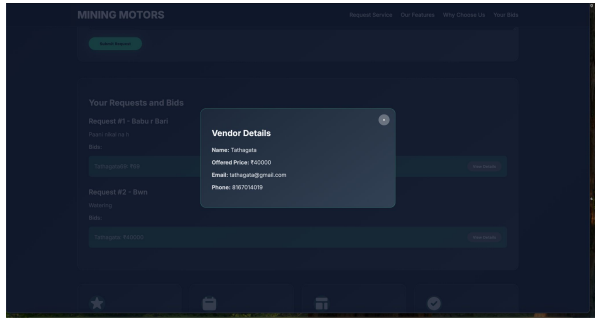
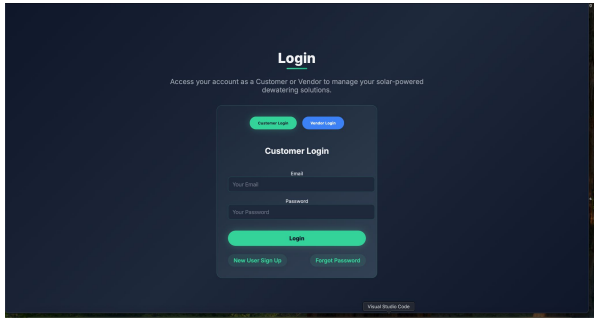
Research:

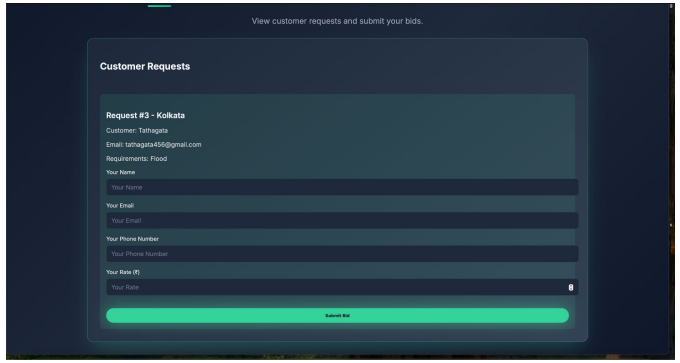
- https://www.sciencedirect.com/science/article/abs/pii/S1364032118300261. (for Solar Panel)
- https://rjwave.org/ijedr/viewpaperforall.php?paper=IJEDR1701064 (for Magnetic Repulsion Engine)
- https://www.xylem.com/en-id/resources/case-studies/a-xylem-dewatering-solution-is-going-strong-at-an-alliance-coal-mine-six-years-after-installation/ (Dewatering practices)
- https://mines.gov.in/admin/download/685a874e549cd1750763342.pdf(Successful Practices)
- https://www.ijmerr.com/v3n1/ijmerr_v3n1_8.pdf
- https://www.nationalgrid.com/stories/energy-explained/how-does-solar-power-work (Efficiency of Solar Power and betterment)



References:







For Experiancing our project

Live Hosted Prototype Portal: https://fancy-gumption-ec4362.netlify.app/

Video for explanation of project: https://youtu.be/YkadWFia1aU